Single cell-level analysis of responses to cancer immunotherapy using a syngeneic murine model of lung cancer

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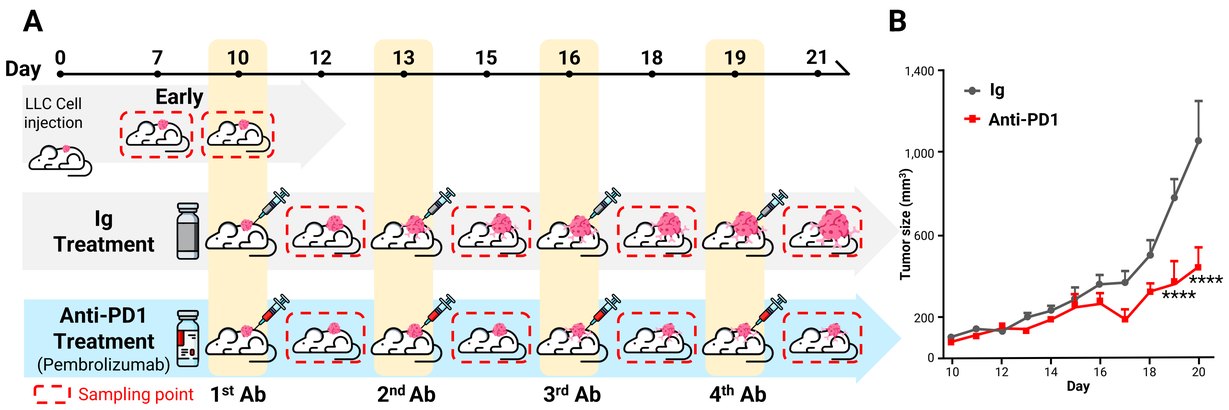
Abstract

Immunotherapy is currently the most promising treatment modality for various types of cancers. It has been demonstrated that targeting immune checkpoints with PD-1 or CTLA-4 blockade (ICB) can lead to long-term survival or complete remission from cancers. However, immunotherapy is limited in efficacy as only a minority of patients benefit from such therapies. Clearly, closer scrutiny of responses from various cells found in tumor tissues during immunotherapy would significantly enhance our understanding of the interactions within the tumor microenvironment (TME) and possibly point to novel targets. Here, we report a longitudinal study using a Lewis lung carcinoma (LLC) syngeneic mouse model and anti-PD-1 antibody treatment. Specifically, we performed single-cell RNA seq using tumor tissues isolated from multiple time points and attempted to reconstruct potentially significant cellular interactions in response to the immunotherapy. Interestingly, natural killer (NK) cells, in particular, showed activation by anti-PD-1 antibody treatment and expressed elevated levels of Xcl-1, which has been implicated in recruiting conventional dendritic cells (cDCs). We present results from gene expression analysis suggesting enhanced antigen-presenting by cDCs and strategies for demonstrating the functionalities of altered cellular responses to immunotherapy.

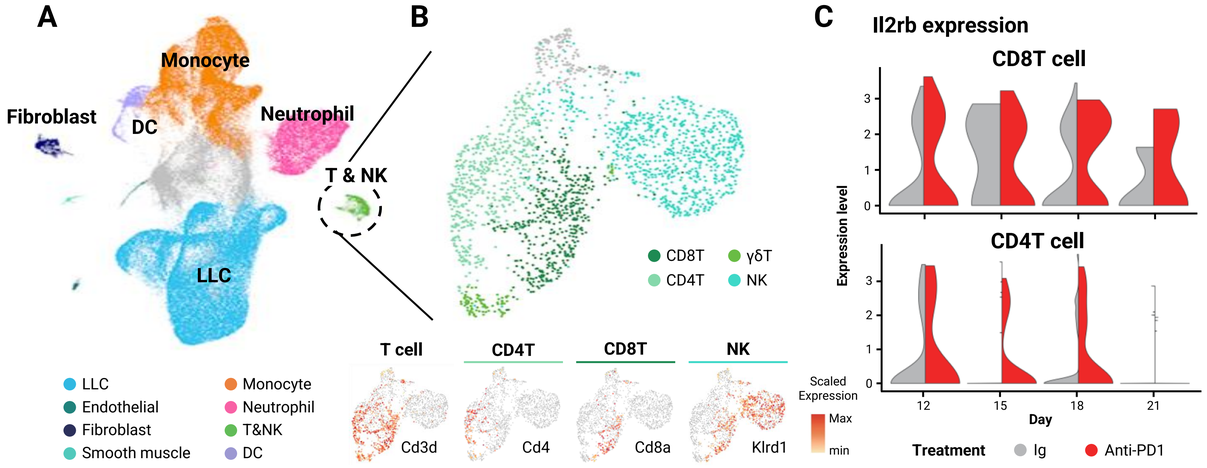
Results

Longitudinal study design using Lewis Lung Carcinoma syngeneic mouse model



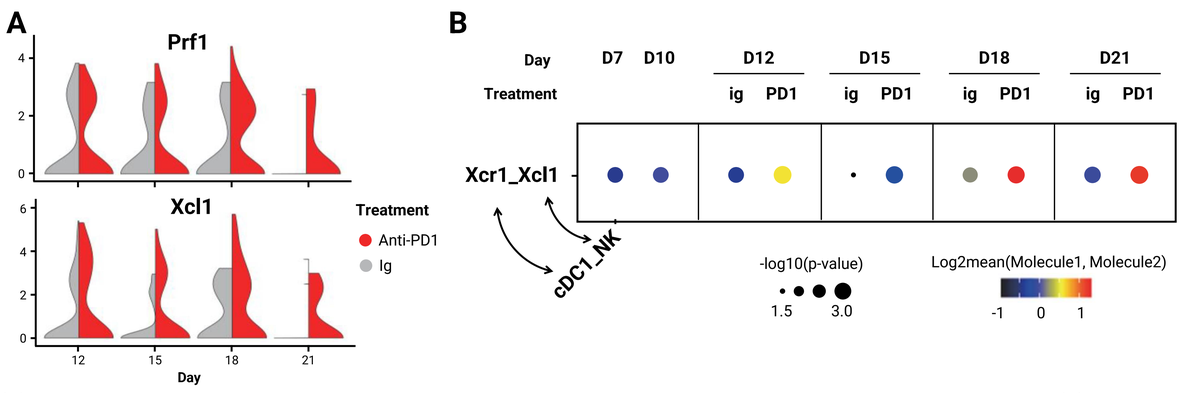
(A) Outline of the longitudinal study. The top black arrow indicates the flow of days, and the day of LLC injection was set as day 0. On days marked with yellow boxes, pembrolizumab was administered, and control mice were simultaneously administered with Ig. (B) Average tumor growth curve of the Ig-treated group and the anti-PD1-treated group. Statistically significant cases are marked with **** ($p < 10^{-4}$ in two-way ANOVA test).

T cells were activated in response to anti-PD1



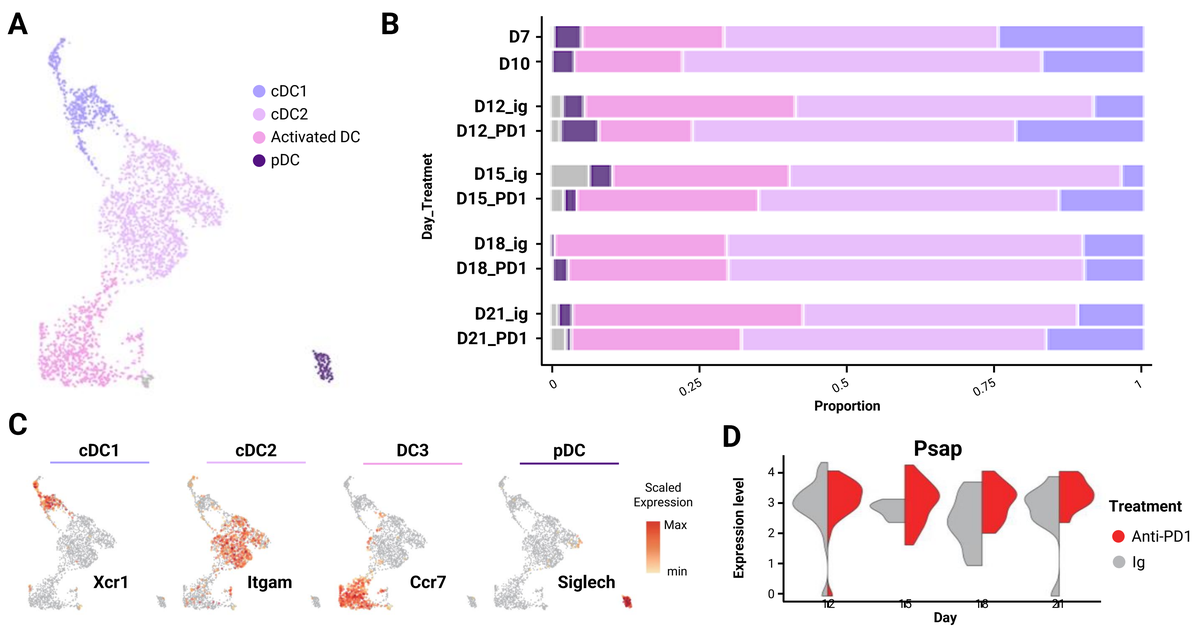
(A) UMAP plot of 124,786 cells in total, annotated into nine major cell types. (B) The T and NK clusters (2,123 cells) were divided into four subgroups, representing CD8T, CD4T, and NK cells. (C) Differential display of *Il2rb* expression (Ig vs. anti-PD1 treatment samples) by day in CD8T and CD4T cells. Il2rb was used as a marker for T cell activation, differentiating CD8+ and CD4+ T cells into effector cells.

NK cell activation and Xcl1 overexpression in anti-PD1 treated samples



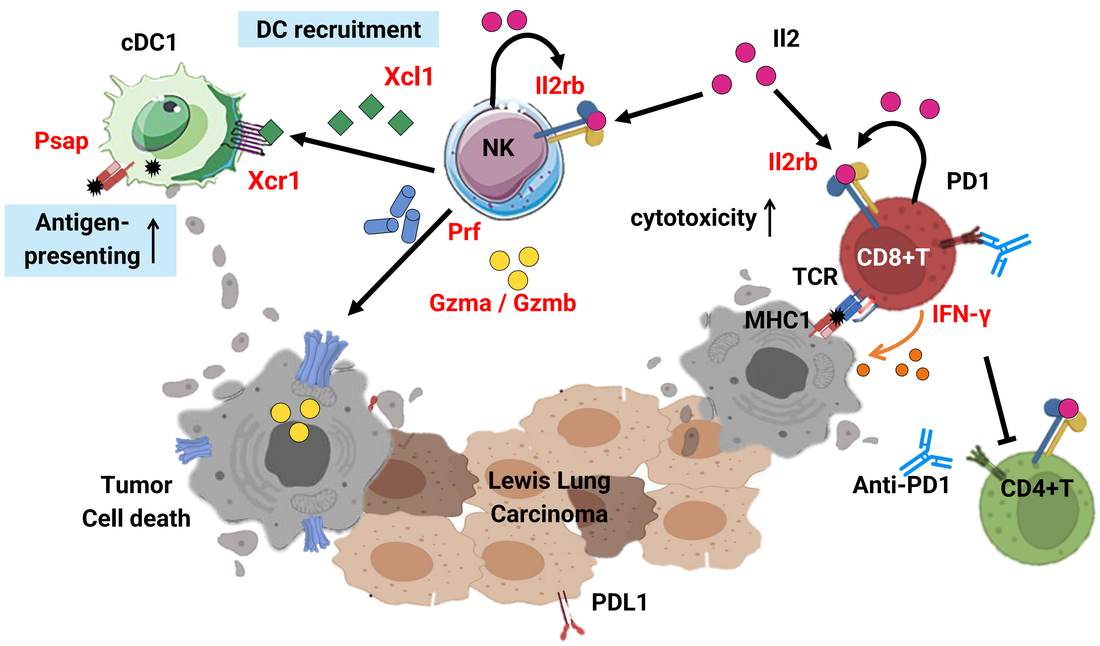
(A) Violin plot shows that Prf1 and Xcl1 are overexpressed on anti-PD1 treatment in NK populations. (B) Cell-cell interaction results predicted by CellPhoneDB. Xcl1 expression in NK cells are strongly correlated with Xcr1 expression in cDC1 cells. The size of the dot is proportional to the significance, and the color of the dot indicates the mean expression of the receptor-ligand complex.

DCs, recruited by NK cells, overexpressed prosaposin (*Psap*)



(A) UMAP colored with the annotated DC subtypes. (B) A bar plot showing the proportions of dendritic cell types for each sample. The ratio of cDC1 increased substantially on anti-PD1 treatment at the initial days (D12 and D15). Colors for cell types are consistent with Fig. (A). (C) Expression of known marker genes for each DC subtype. (D) Violin plot shows overexpression of *Psap* on anti-PD1 treatment in cDC1 cells. Psap is an essential molecule for processing apoptotic body and presenting lipid antigens in DCs.

Conclusion



Summary overview of immune cell responses to anti-PD1 treatment.

From the differentially expressed genes analysis of single-cell transcriptome data, we showed that T cells and NK were activated on ICB treatment. Specifically, NK cells overexpressed Xcl1 and recruited cDC1. We hypothesize that the antigen-presenting ability of DC was enhanced via increased Psap expression. Validation experiments are currently in progress.